CX-4 Capper Maintenance Manual

Overview

The CX-4 is a four-head rotary capper that applies screw caps to bottles on the Riverbend Packaging filling line. Each head uses a magnetic clutch to limit applied torque. Maintenance personnel calibrate the clutches, change over the heads for new cap sizes, and perform routine upkeep as described in this manual.

Specifications

Item	Value
Model	CX-4
Heads	4, magnetic clutch
Speed	Up to 120 bottles per minute
Cap sizes	28 mm to 38 mm
Torque range (clutch)	5 to 40 in-lb
Air supply	80 psi, filtered and dry

Torque Head Calibration

Torque heads must be calibrated at every changeover and after any head repair. Use a calibrated digital torque tester with a current certificate.

Calibration Procedure

1. Stop the capper and perform lockout/tagout on disconnect MD-7 and air valve AV-2.
2. Remove the guard panel above the heads.
3. Mount the torque tester adapter on the first head.
4. Restore power temporarily using the maintenance procedure and cycle the head against the tester.
5. Adjust the clutch collar until the tester reads within the target range below.
6. Record the reading for each head on the calibration sheet.
7. Repeat for all four heads. Spread between heads must not exceed 1.5 in-lb.
8. Replace guard panels and remove locks.

Calibration Targets

Set the heads to **15 to 20 in-lb** for 38 mm standard closures on the 500 mL bottle. Record the setting on the calibration sheet and tag the head with the date.

Cap size	Bottle	Calibration setting
38 mm	500 mL	15 to 20 in-lb
28 mm	250 mL	10 to 14 in-lb
33 mm	1 L	14 to 18 in-lb

NOTE: If a head drifts more than 2 in-lb between calibrations, replace the clutch spring.

Changeover

1. Run remaining product through the line and stop the capper.
2. Lock out electrical and pneumatic energy.
3. Remove the chuck inserts and star wheel for the old size.
4. Install the matching chuck inserts, star wheel, and chute guides for the new cap size.
5. Adjust the capper height using the hand wheel so the chuck touches the cap top without pressing.
6. Perform torque calibration for the new size.
7. Remove locks, restore power, and run ten test bottles.
8. Check cap application visually and release the line to Quality for first-article inspection.

Troubleshooting

Symptom	Likely cause	Corrective action
Cap cocked or crossed	Chuck insert worn or height misadjusted	Replace insert; reset capper height
Cap torque drifting	Worn clutch spring	Replace spring; recalibrate
Caps missing at infeed	Chute obstruction	Clear chute after lockout
Head will not rotate	Pneumatic fault or seized bearing	Check air pressure; inspect bearing
Scuffed caps	Chuck contamination	Clean chuck inserts
Bottle spinning during capping	Worn neck grippers	Replace grippers

Preventive Maintenance

Interval	Task
Daily	Wipe chuck inserts; check air pressure
Weekly	Lubricate head bearings with food-grade grease
Monthly	Inspect clutch springs and grippers
Quarterly	Verify torque calibration against certified tester
Annually	Rebuild heads; certify torque tester

Lubrication

Use only NSF H1 food-grade grease. Over-greasing contaminates caps and bottles. Apply two pumps to each zerk fitting and wipe excess.

Safety Notes

Guard panels must be in place during operation. Do not reach into the star wheel area while the capper is energized. Follow the Lockout/Tagout Procedure for any servicing inside the guarded area.

Documentation

Calibration sheets are retained in the maintenance binder for one year and copies provided to Quality on request. Report any calibration out-of-tolerance finding to the Maintenance Supervisor.